

Neural network



Neural Network: without the brain stuff

(Before) Linear score function: $f = Wx$

(Now) 2-layer Neural Network: $f = W_2 \max(0, W_1 x)$



INT305 Machine Learning
Lecture 5

Neural Network and Back Propagation

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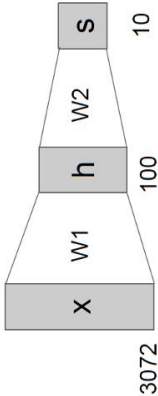
Neural network



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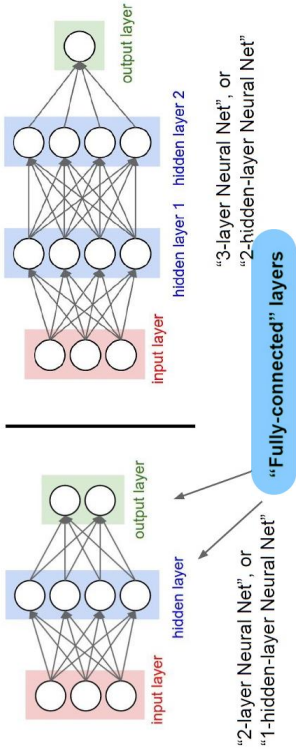


Neural Network: without the brain stuff 不涉及大脑相关内容

(Before) Linear score function: $f = Wx$

Neural network

Neural Networks: Architectures



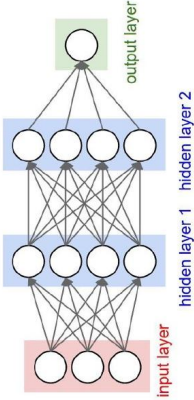
Neural network

Neural Network: without the brain stuff

(Before) Linear score function: $f = Wx$
(Now) 2-layer Neural Network: $f = W_2 \max(0, W_1 x)$
or 3-layer Neural Network:
 $f = W_3 \max(0, W_2 \max(0, W_1 x))$

Neural network

Example Feed-forward computation of a Neural Network



```
# forward-pass of a 3-layer neural network:  
f = Lambda x: 1.0/(1.0 + np.exp(-x)) # activation function (use sigmoid)  
x = np.random.randn(3, 1) # random input vector of three numbers (3x1)  
h1 = f(np.dot(W1, x) + b1) # calculate first hidden layer activations (4x1)  
h2 = f(np.dot(W2, h1) + b2) # calculate second hidden layer activations (4x1)  
out = np.dot(W3, h2) + b3 # output neuron (1x1)
```

Activation functions

Activation Functions

Sigmoid
 $\sigma(x) = 1/(1 + e^{-x})$

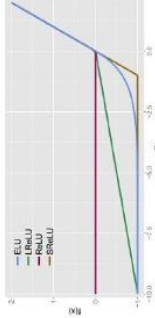
tanh $\tanh(x)$

ReLU $\max(0, x)$



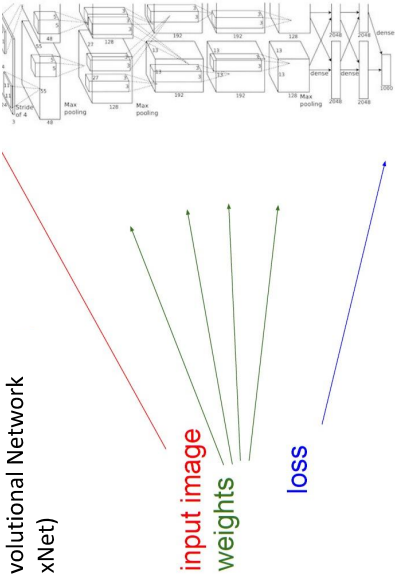
Maxout $\max(w_1^T x + b_1, w_2^T x + b_2)$

ELU $f(x) = \begin{cases} x & \text{if } x > 0 \\ \alpha(\exp(x) - 1) & \text{if } x \leq 0 \end{cases}$



Computational Graph

Convolutional Network
(AlexNet)



Gradient Descent

Where we are...

$$s = f(x; W) = Wx$$

scores function

$$L_i = \sum_{j \neq y_i} \max(0, s_j - s_{y_i} + 1)$$

SVM loss

data loss + regularization

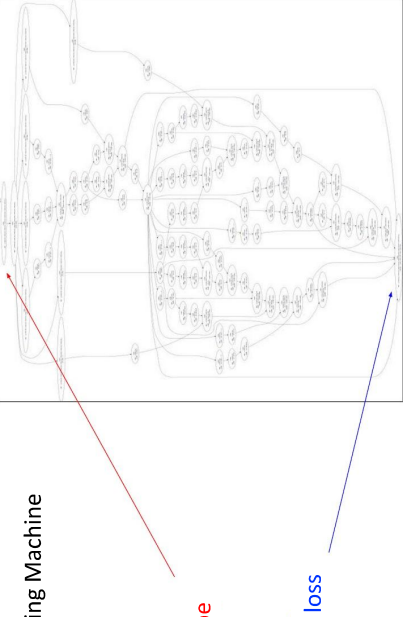
$$L = \frac{1}{N} \sum_{i=1}^N L_i + \sum_k W_k^2$$

want

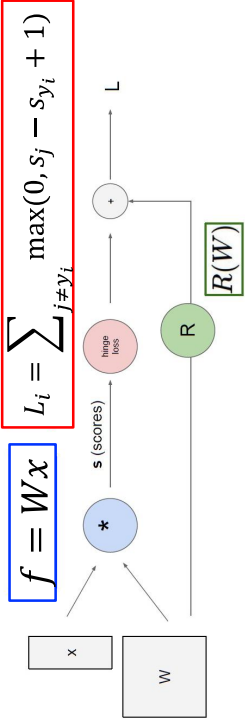
$$\nabla_W L$$

Computational Graph

Neural Turing Machine



Computational Graph



Example 1

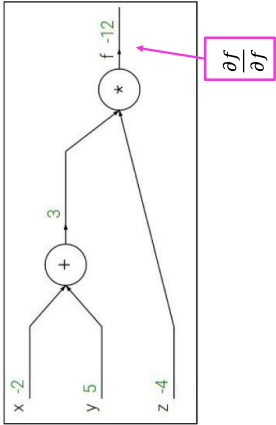
$f(x,y,z) = (x + y)z$

e.g. $x = -2, y = 5, z = -4$

$q = x + y \quad \frac{\partial q}{\partial x} = 1, \frac{\partial q}{\partial y} = 1$

$f = qz \quad \frac{\partial f}{\partial q} = z, \frac{\partial f}{\partial z} = q$

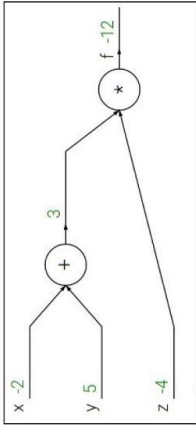
Want: $\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}, \frac{\partial f}{\partial z}$



Example 1

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Example 1

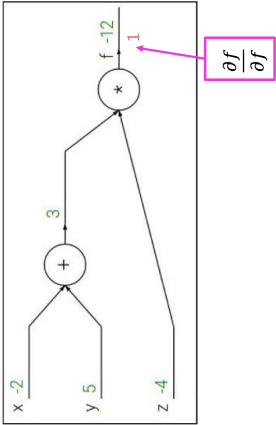
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Want: $\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}, \frac{\partial f}{\partial z}$



$\frac{\partial f}{\partial x} = \frac{\partial f}{\partial q} \frac{\partial q}{\partial x} = z \cdot 1 = -4$
 $\frac{\partial f}{\partial y} = \frac{\partial f}{\partial q} \frac{\partial q}{\partial y} = z \cdot 1 = -4$
 $\frac{\partial f}{\partial z} = q = 3$

Example 1

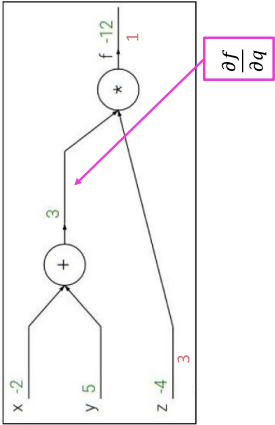
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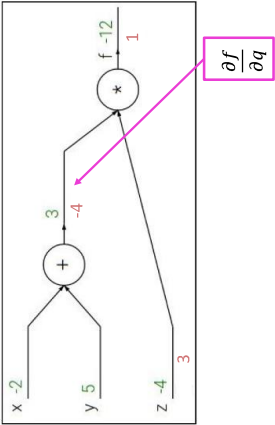
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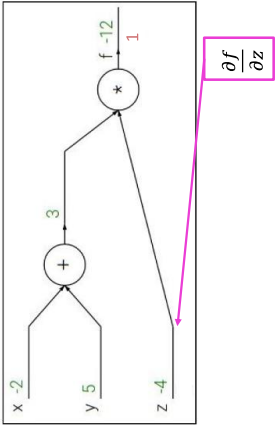
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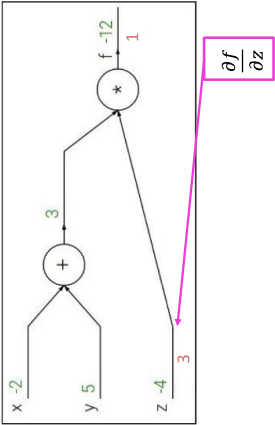
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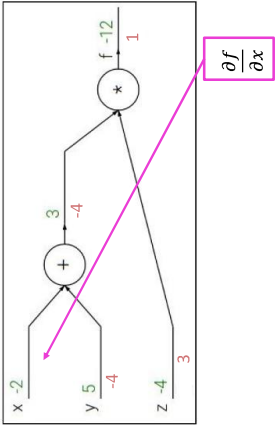
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$\frac{\partial f}{\partial x}$

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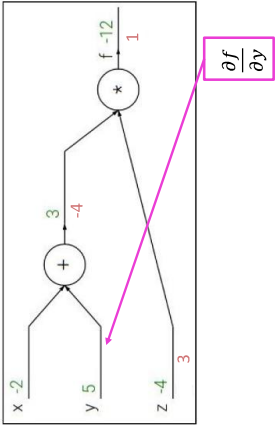
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$\frac{\partial f}{\partial y}$

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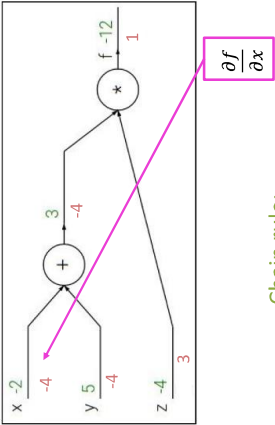
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Want: $\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}, \frac{\partial f}{\partial z}$



$\frac{\partial f}{\partial x}$

Chain rule:

$$\frac{\partial f}{\partial y} = \frac{\partial f}{\partial q} \frac{\partial q}{\partial y}$$

Example 1

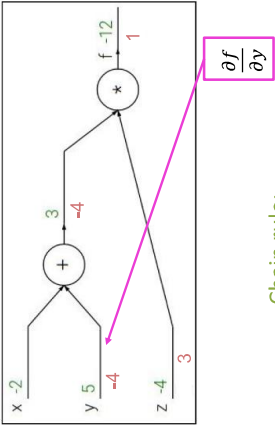
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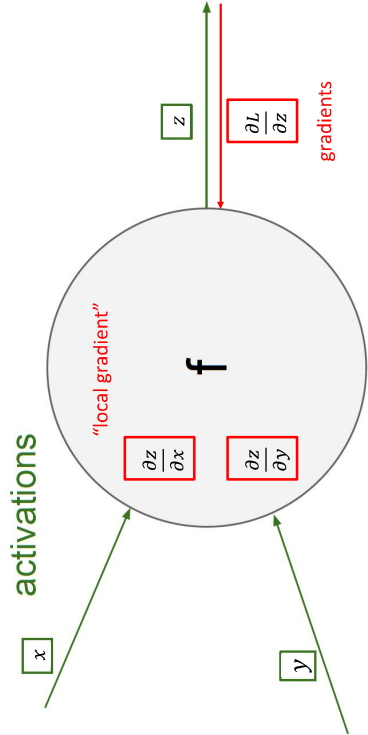
$\frac{\partial f}{\partial y}$

Chain rule:

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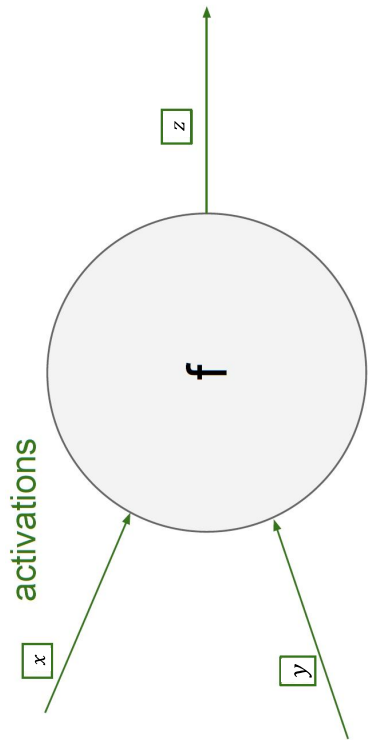
X

Chain rule



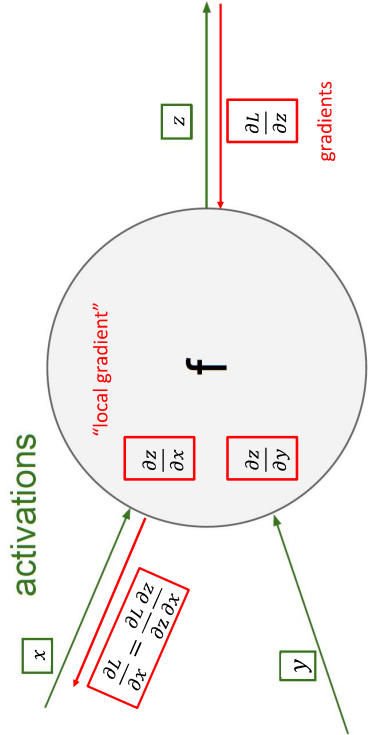
X

Chain rule



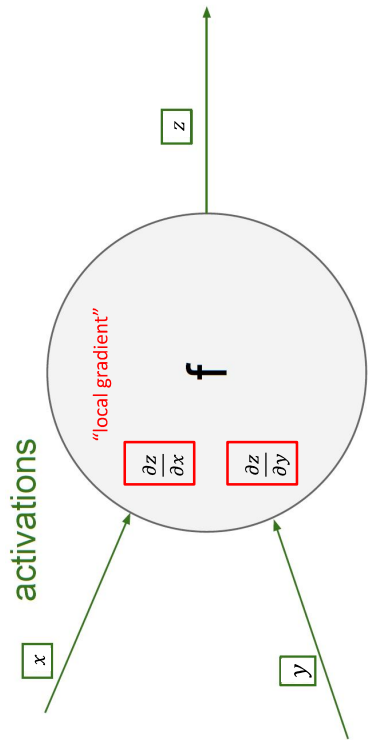
X

Chain rule



X

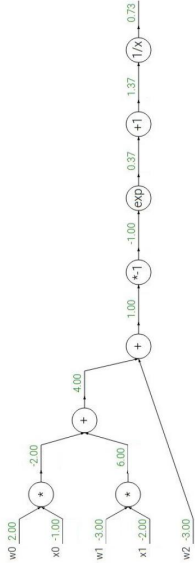
Chain rule



Chain rule

Another example:

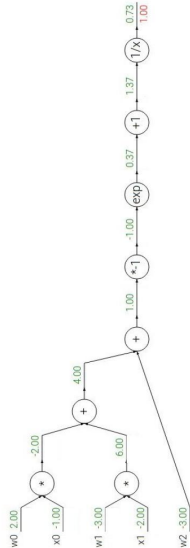
$$f(w, x) = \frac{1}{1 + e^{-(w_0x_0 + w_1x_1 + w_2)}} \quad$$



Chain rule

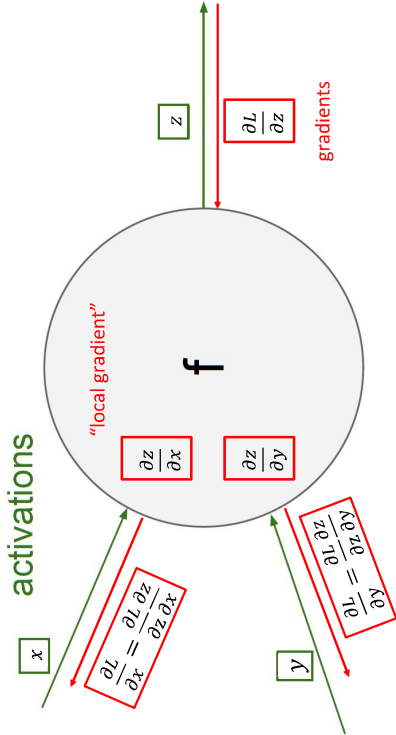
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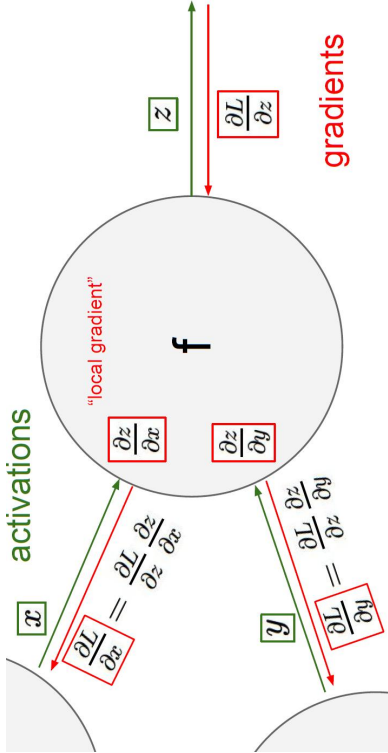


$f(x) = e^x$	$\rightarrow$	$\frac{df}{dx} = e^x$	$\left \right.$	$f(x) = \frac{1}{x}$	$\rightarrow$	$\frac{df}{dx} = -1/x^2$
$f_a(x) = ax$	$\rightarrow$	$\frac{df}{dx} = a$	$\left \right.$	$f_c(x) = c + x$	$\rightarrow$	$\frac{df}{dx} = 1$

Chain rule

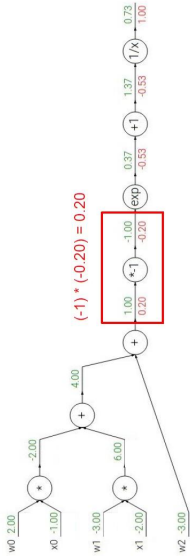


Chain rule



Chain rule

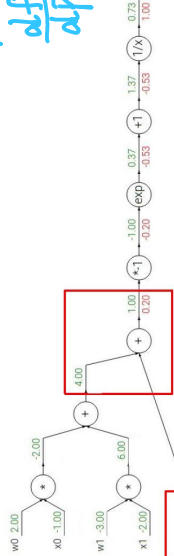
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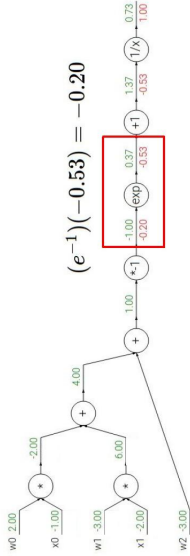


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$p + w_2 = 1$
 $\frac{df}{dp} = \frac{df}{dz} \times \frac{dz}{dp}$
 $=$

Chain rule

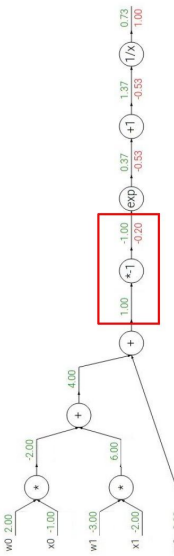
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Chain rule

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[local gradient] x [its gradient]
x0: [2] x [0.2] = 0.4
w0: [-1] x [0.2] = -0.2

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Sigmoid

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Chain rule

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[local gradient] x [its gradient]
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[1] x [0.2] = 0.2 (both inputs!)

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Exercise 1

Pooling units take n values x_i , $i \in [1, n]$ and compute a scalar output whose value is invariant to permutations of the inputs.

1. The Lp-pooling module takes positive inputs and computes

$y = (\sum_i x_i^p)^{\frac{1}{p}}$, assuming we know that $y' = \frac{\partial L}{\partial y}$, what is $x_i' = \frac{\partial L}{\partial x_i}$?

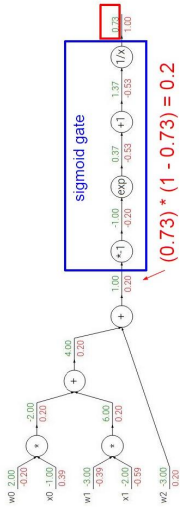
$\frac{\partial L}{\partial x_i} = \frac{\partial L}{\partial y} \times \frac{\partial y}{\partial x_i} = y' \cdot \frac{\partial}{\partial x_i} (\sum_i x_i^p)^{\frac{1}{p}} = y' \cdot \frac{1}{p} (\sum_i x_i^p)^{\frac{1}{p}-1} \cdot p x_i^{p-1} = y' \cdot \frac{x_i^{p-1}}{(\sum_i x_i^p)^{\frac{p-1}{p}}}$

2. The log-average module computes $y = \frac{1}{\beta} \ln(\sum_i \exp(\beta x_i))$, assuming we know that $y' = \frac{\partial L}{\partial y}$, what is $x_i' = \frac{\partial L}{\partial x_i}$?

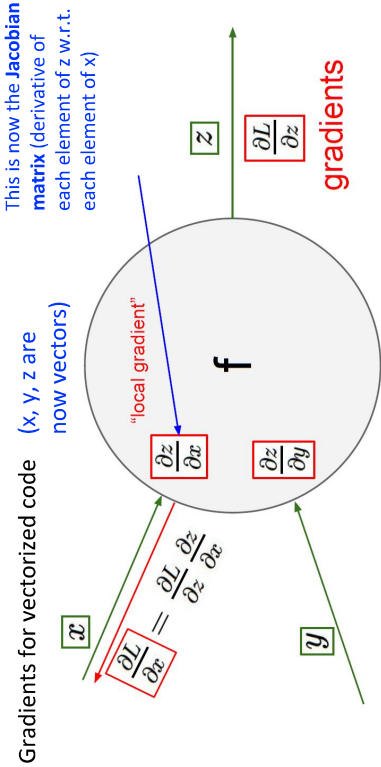
Sigmoid

$f(w, x) = \frac{1}{1 + e^{-(w_0 x_0 + w_1 x_1 + w_2)}} \quad \sigma(x) = \frac{1}{1 + e^{-x}} \quad \text{sigmoid function}$

$\frac{d\sigma(x)}{dx} = \frac{e^{-x}}{(1 + e^{-x})^2} = \left(\frac{1 + e^{-x} - 1}{1 + e^{-x}} \right) \left(\frac{1}{1 + e^{-x}} \right) = (1 - \sigma(x)) \sigma(x)$

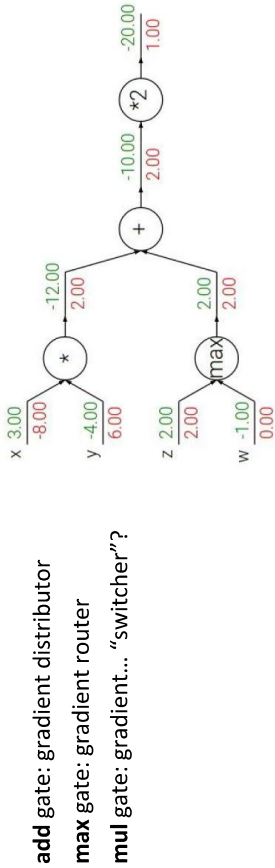


Gradients for vector



Pattern in backward flow

Patterns in backward flow

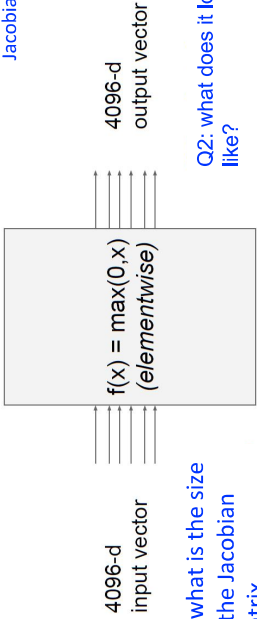


Gradients for vector

Vectorized operation

$$\frac{\partial L}{\partial x} = \boxed{\frac{\partial f}{\partial x} \frac{\partial L}{\partial f}}$$

Jacobian matrix

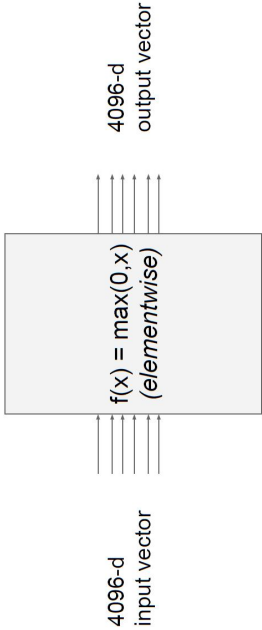


Q: what is the size of the Jacobian matrix
[4096 x 4096!]

Q2: what does it look like?

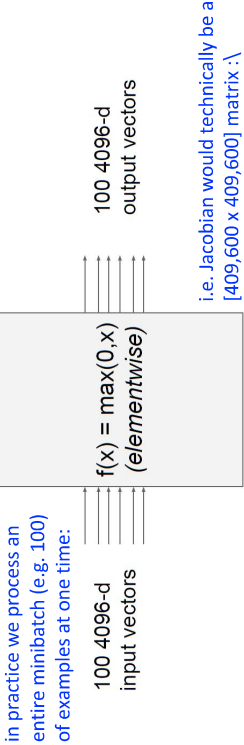
Gradients for vector

Vectorized operation



Gradients for vector

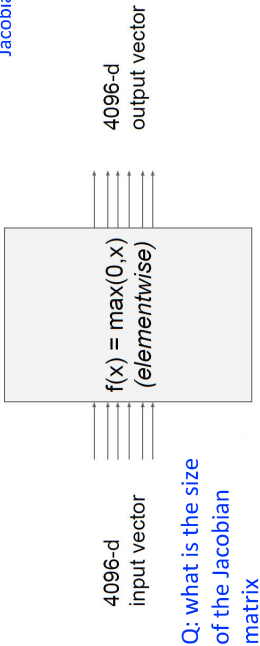
Vectorized operation



Vectorized operation

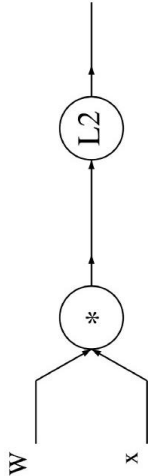
$$\frac{\partial L}{\partial x} = \boxed{\frac{\partial f}{\partial x} \frac{\partial L}{\partial f}}$$

Jacobian matrix



Gradients for vector

A vectorized example: $f(x, W) = \|W \cdot x\|^2 = \sum_{i=1}^n (W \cdot x)_i^2$



Gradients for vector

A vectorized example: $f(x, W) = \|W \cdot x\|^2 = \sum_{i=1}^n (W \cdot x)_i^2$

$\begin{bmatrix} 0.1 & 0.5 \\ -0.3 & 0.8 \end{bmatrix}$

$\begin{bmatrix} 0.2 \\ 0.4 \end{bmatrix}$

A computational graph showing the calculation of the squared norm of the product of W and x. The inputs W and x are connected to a multiplication node (*). The output of the multiplication node is connected to a squaring node (L2).

$$q = W \cdot x = \begin{pmatrix} W_{1,1}x_1 + \dots + W_{1,n}x_n \\ \vdots \\ W_{n,1}x_1 + \dots + W_{n,n}x_n \end{pmatrix}$$

$$f(q) = \|q\|^2 = q_1^2 + \dots + q_n^2$$

Gradients for vector

A vectorized example: $f(x, W) = \|W \cdot x\|^2 = \sum_{i=1}^n (W \cdot x)_i^2$

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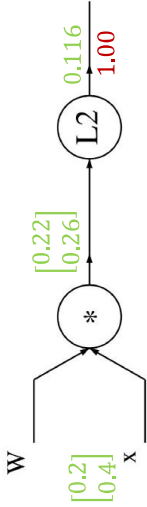
$\Downarrow$
 $\in \mathbb{R}^n \in \mathbb{R}^{n \times n}$

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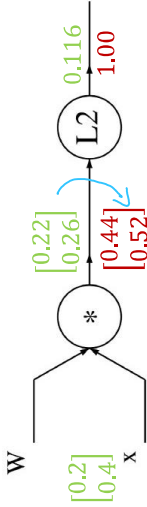
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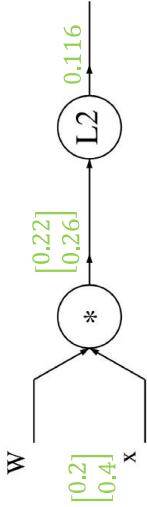
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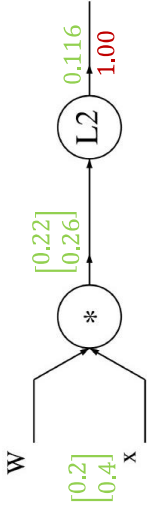
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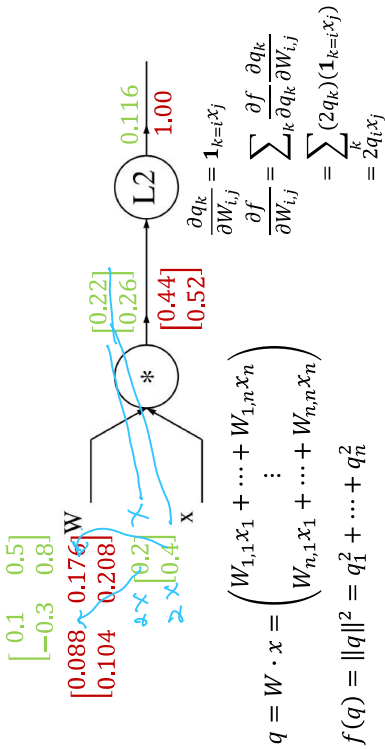


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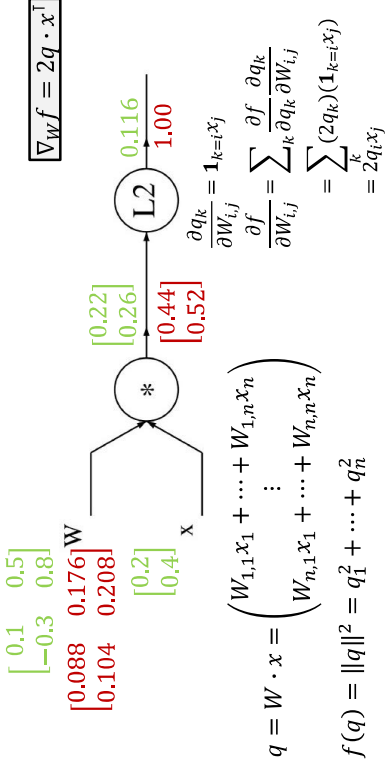
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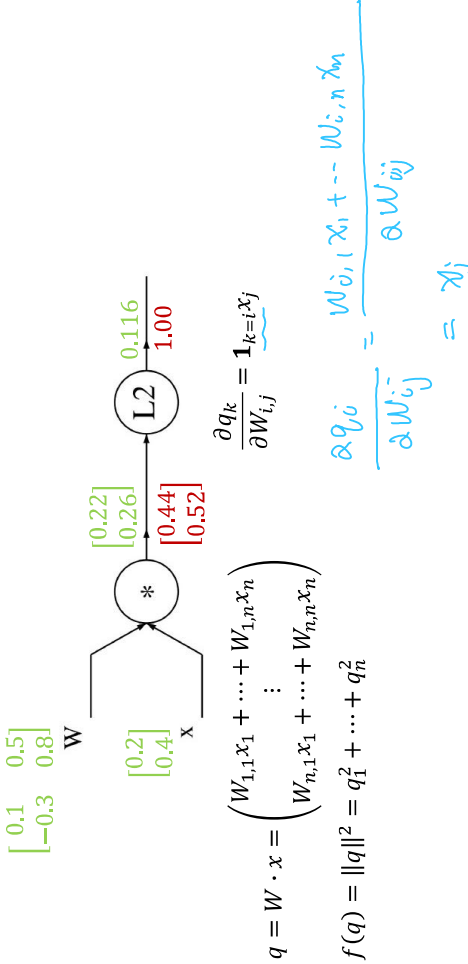
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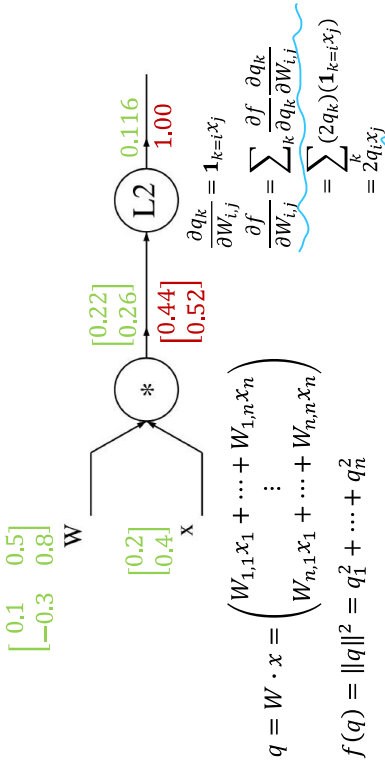
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$\begin{bmatrix} 0.22 \\ 0.26 \end{bmatrix}$

$\begin{bmatrix} 0.2 \\ 0.4 \end{bmatrix}$

x

$*$

$\begin{bmatrix} 0.44 \\ 0.52 \end{bmatrix}$

$\begin{bmatrix} 0.116 \\ 1.00 \end{bmatrix}$

$L2$

$\frac{\partial q_k}{\partial x_i} = W_{k,i}$

$\frac{\partial f}{\partial x_i} = \sum_k \frac{\partial f}{\partial q_k} \frac{\partial q_k}{\partial x_i}$

$= \sum_k 2q_k W_{k,i}$

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$q = Wx$

Always check: The gradient with respect to a variable should have the same shape as the variable

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